GPU Batch Lunar Gravity Validation

```
Scenarios      : 100
Seed           : 42
Altitude range : 200-400 km
Duration       : 1 days
RK4 dt         : 10 s
Output dt      : 60 s
Dtype          : float64
Truth          : SH200 DOP853
GPU models     : st_lrps
Frame mode     : match_dynamics_engine

SH200 DOP853 is the numerical reference, not physical truth.
GPU SH200 RK4 quantifies fixed-step/framework error against that reference.
```

Executive Summary

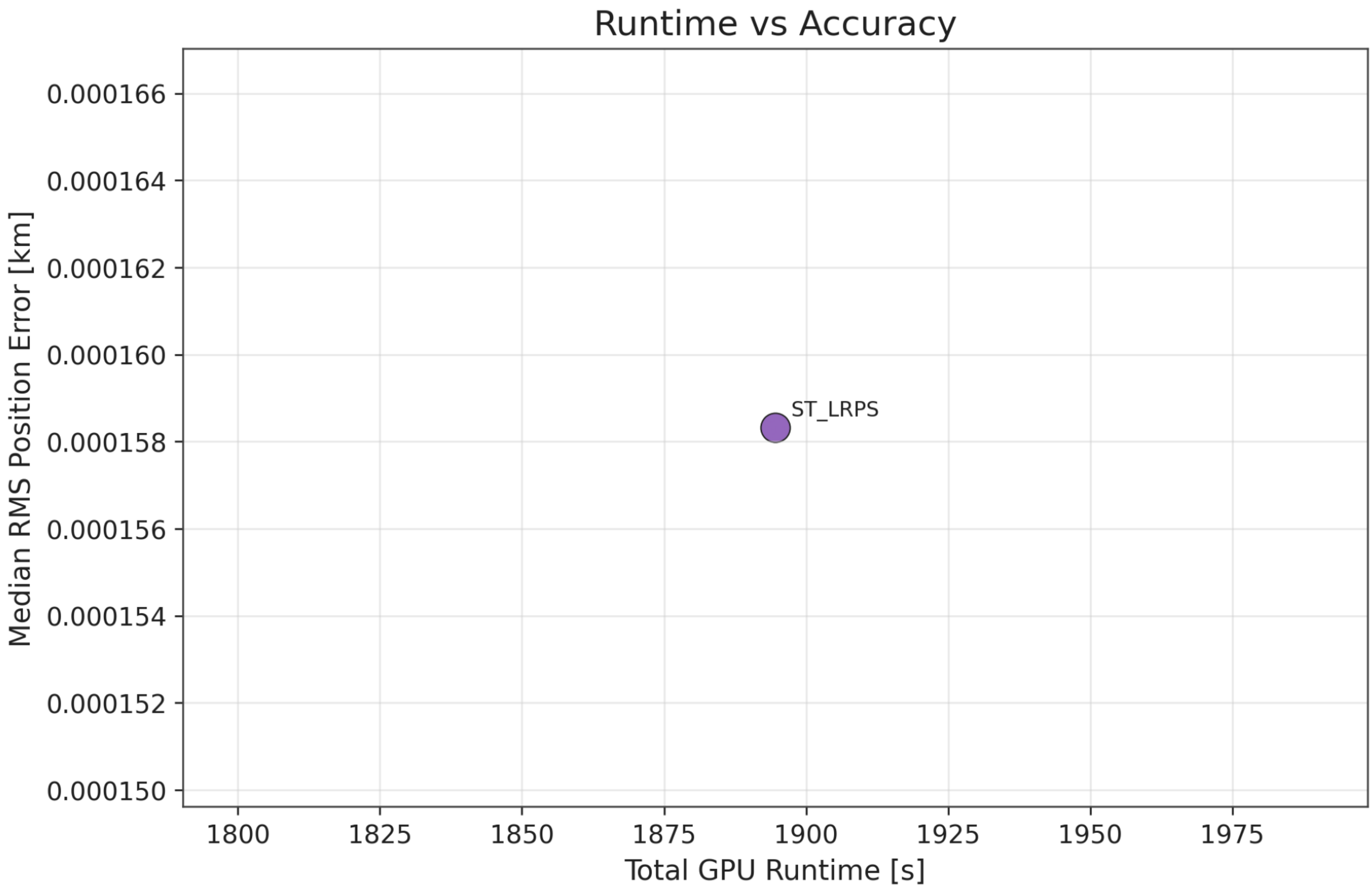
```
Most accurate GPU model : GPU_ST_LRPS_RK4
Fastest GPU model       : GPU_ST_LRPS_RK4
ST-LRPS closest by median RMS : n/a
ST-LRPS median-equivalent status: insufficient_data
ST-LRPS closest by P95 RMS    : n/a

Best ST-LRPS scenario          : 55
Representative ST-LRPS scenario: 81
Worst ST-LRPS scenario         : 36
```

Accuracy Ranking Table

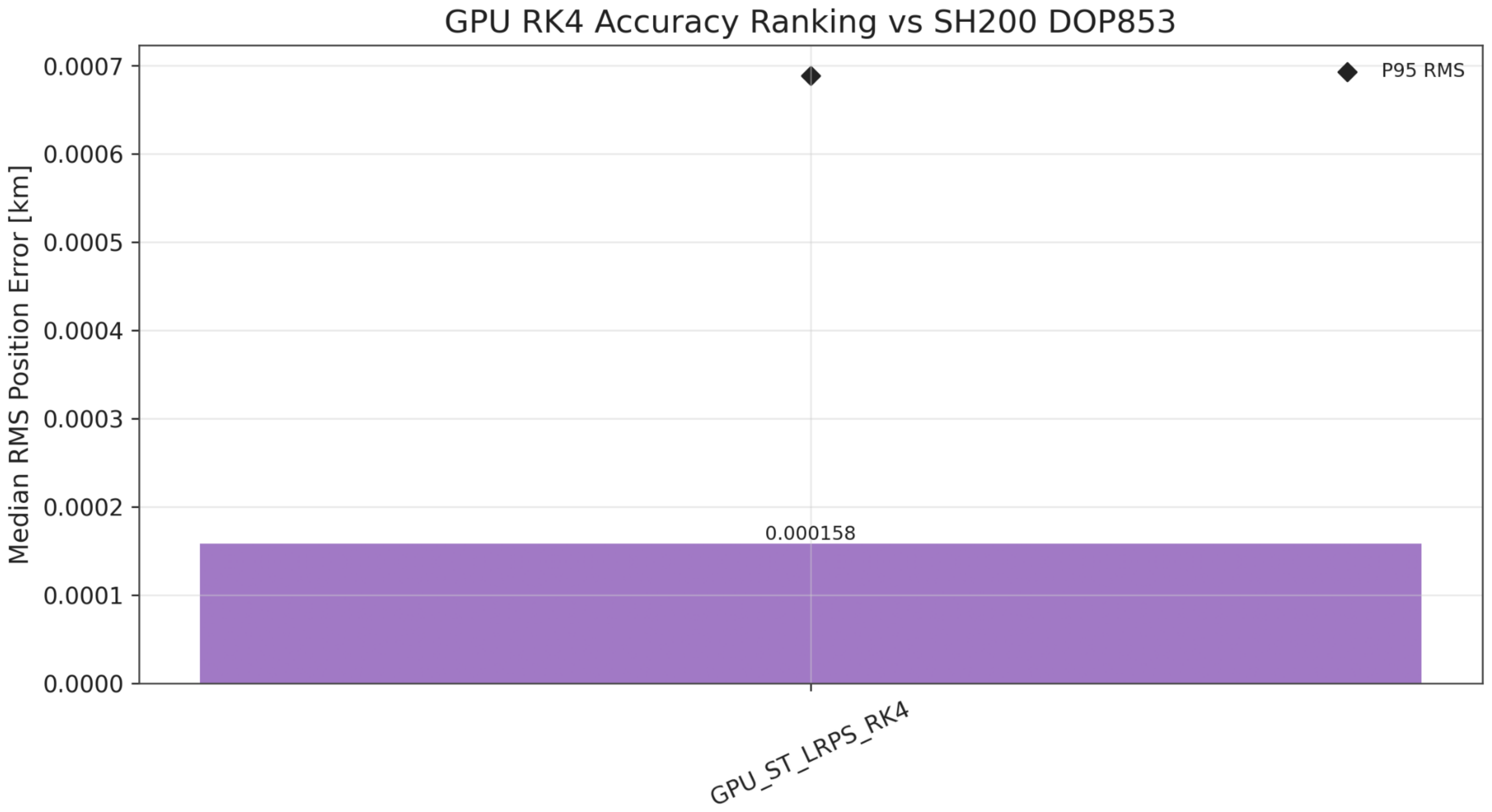
Model	Median RMS	P95 RMS	Max RMS	Median Along
GPU_ST_LRPS_RK4	0.0002	0.0007	0.0010	0.0002

Runtime Table

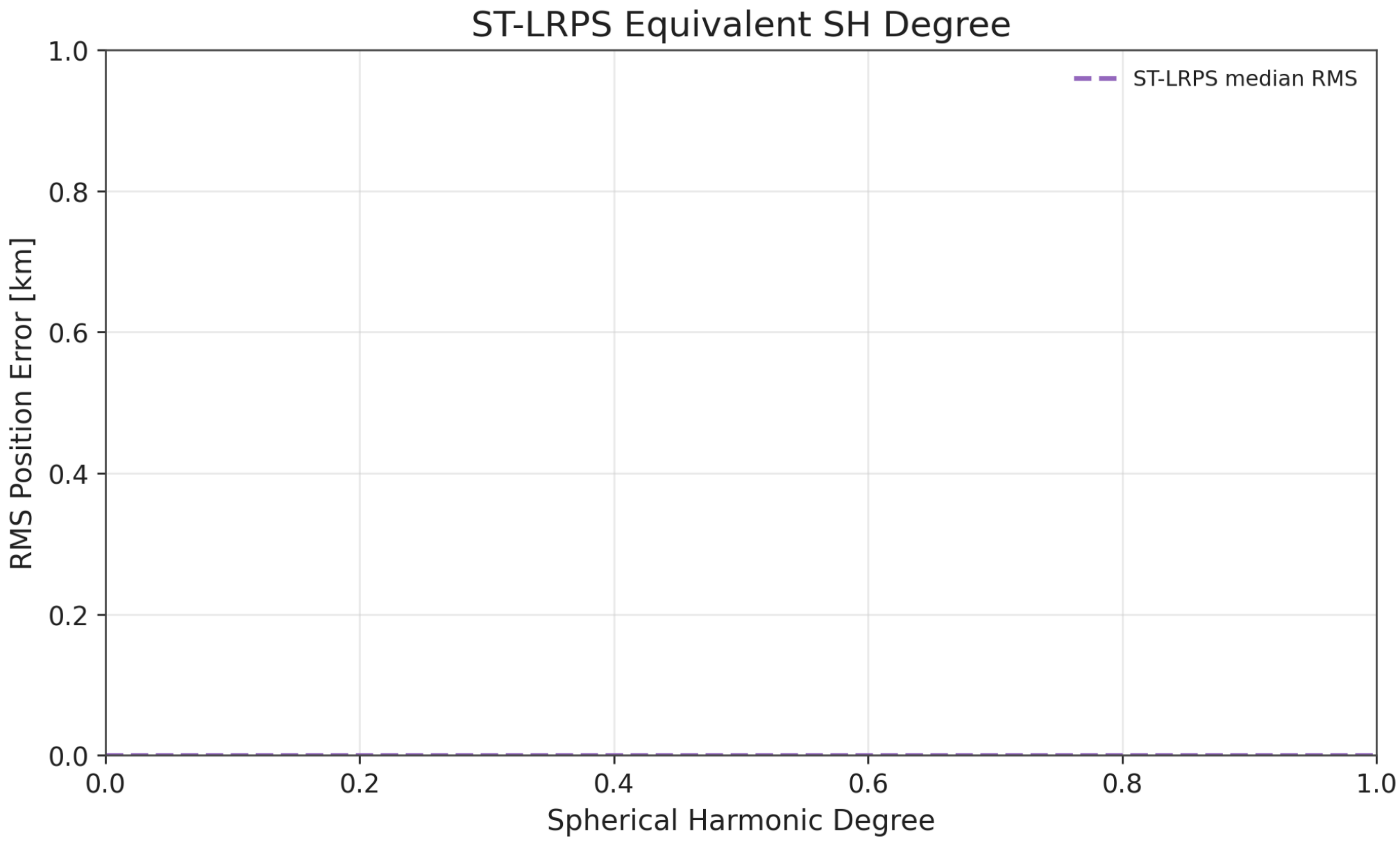
Model	Runtime s	Runtime/sc	Traj-steps/s	Speedup truth
GPU_ST_LRPS_RK4	1894.571	18.94571	456.0	2.25



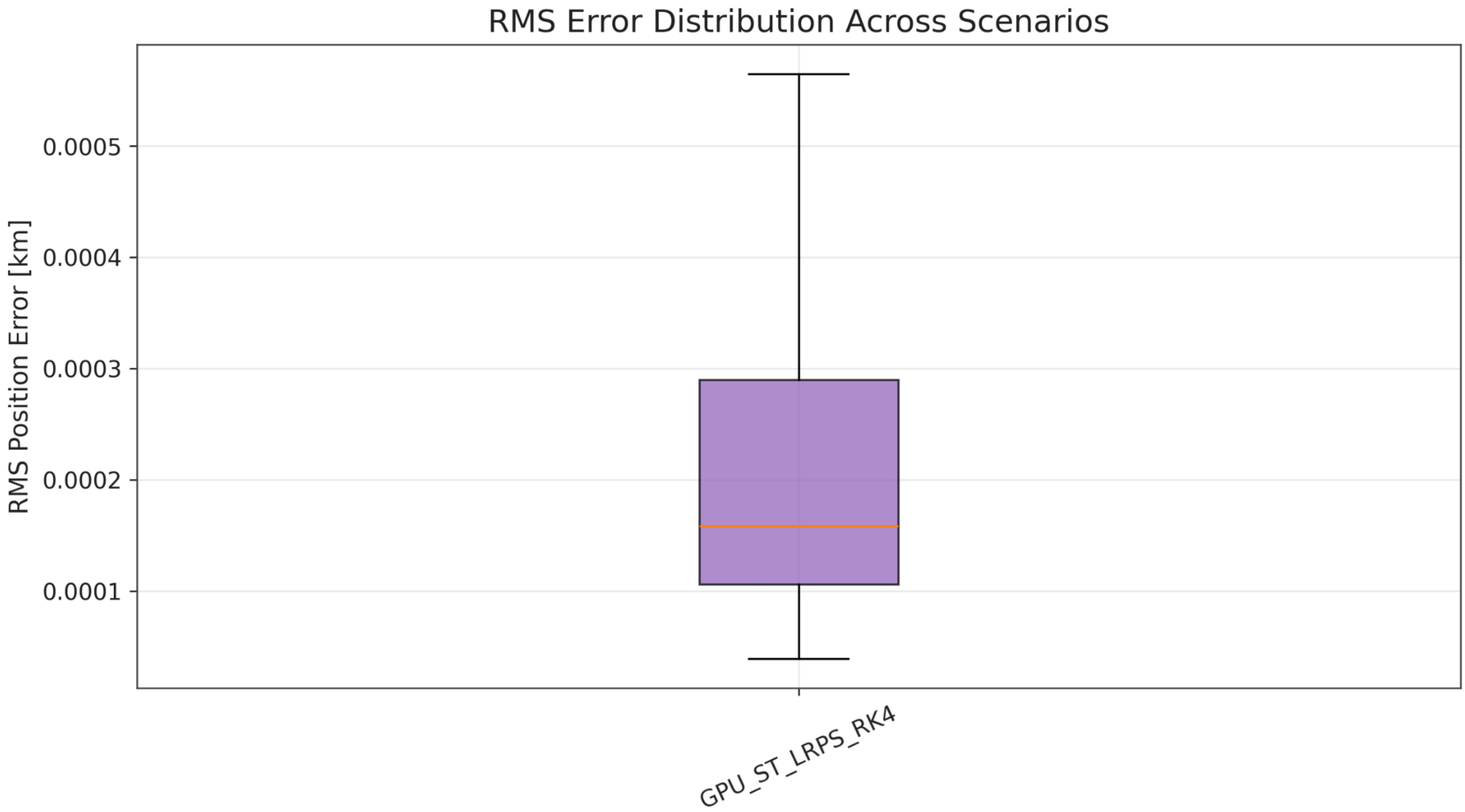
Runtime vs Accuracy
Median RMS Position Error [km]
0.000166
0.000164
0.000162
0.000160
0.000158
0.000156
0.000154
0.000152
0.000150
ST_LRPS
1800
1825
1850
1875
1900
1925
1950
1975
Total GPU Runtime [s]

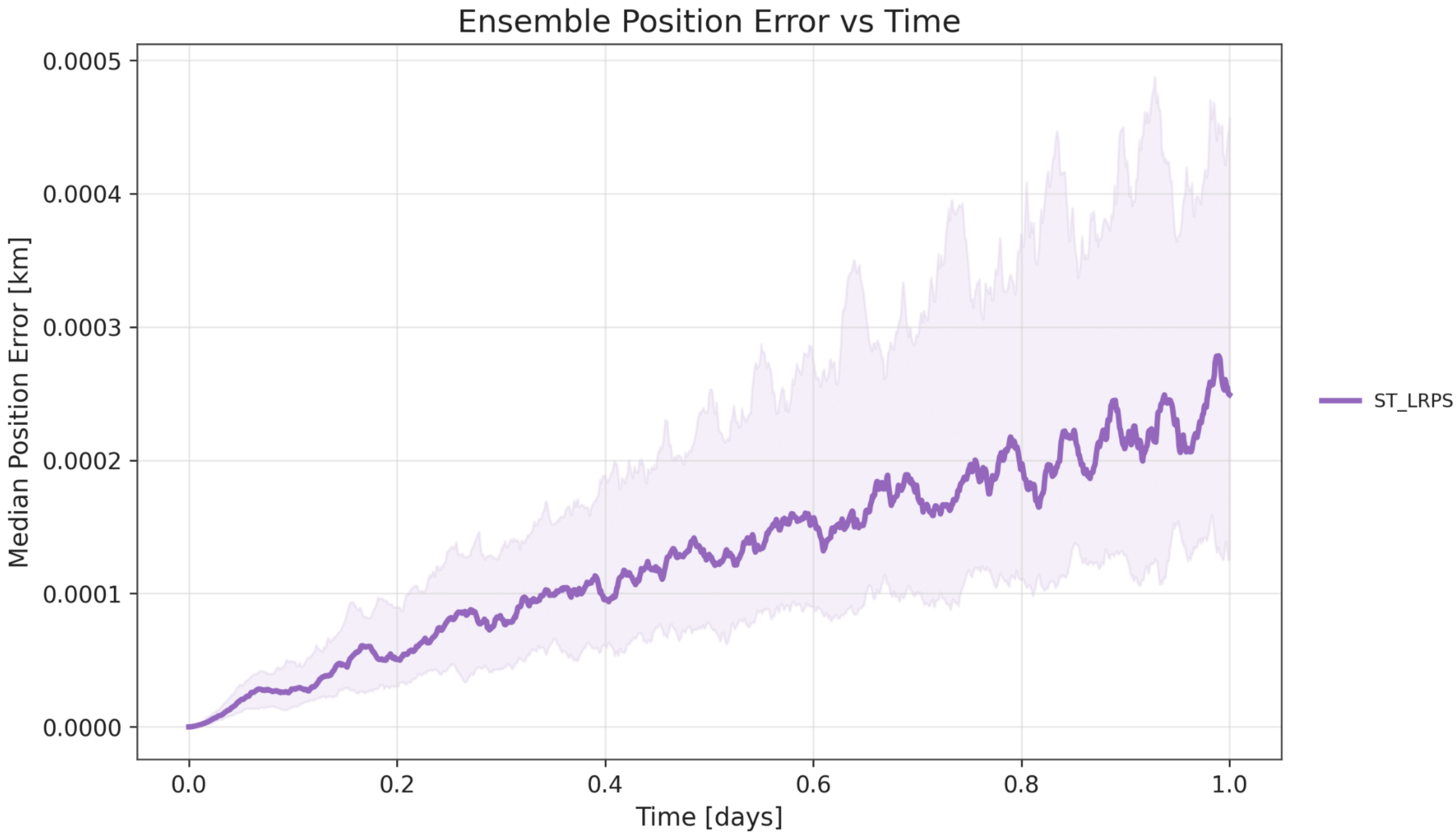


GPU RK4 Accuracy Ranking vs SH200 DOP853
Median RMS Position Error [km]
0.0007
0.0006
0.0005
0.0004
0.0003
0.0002
0.0001
0.0000
P95 RMS
0.000158
GPU_ST_LRPS_RK4

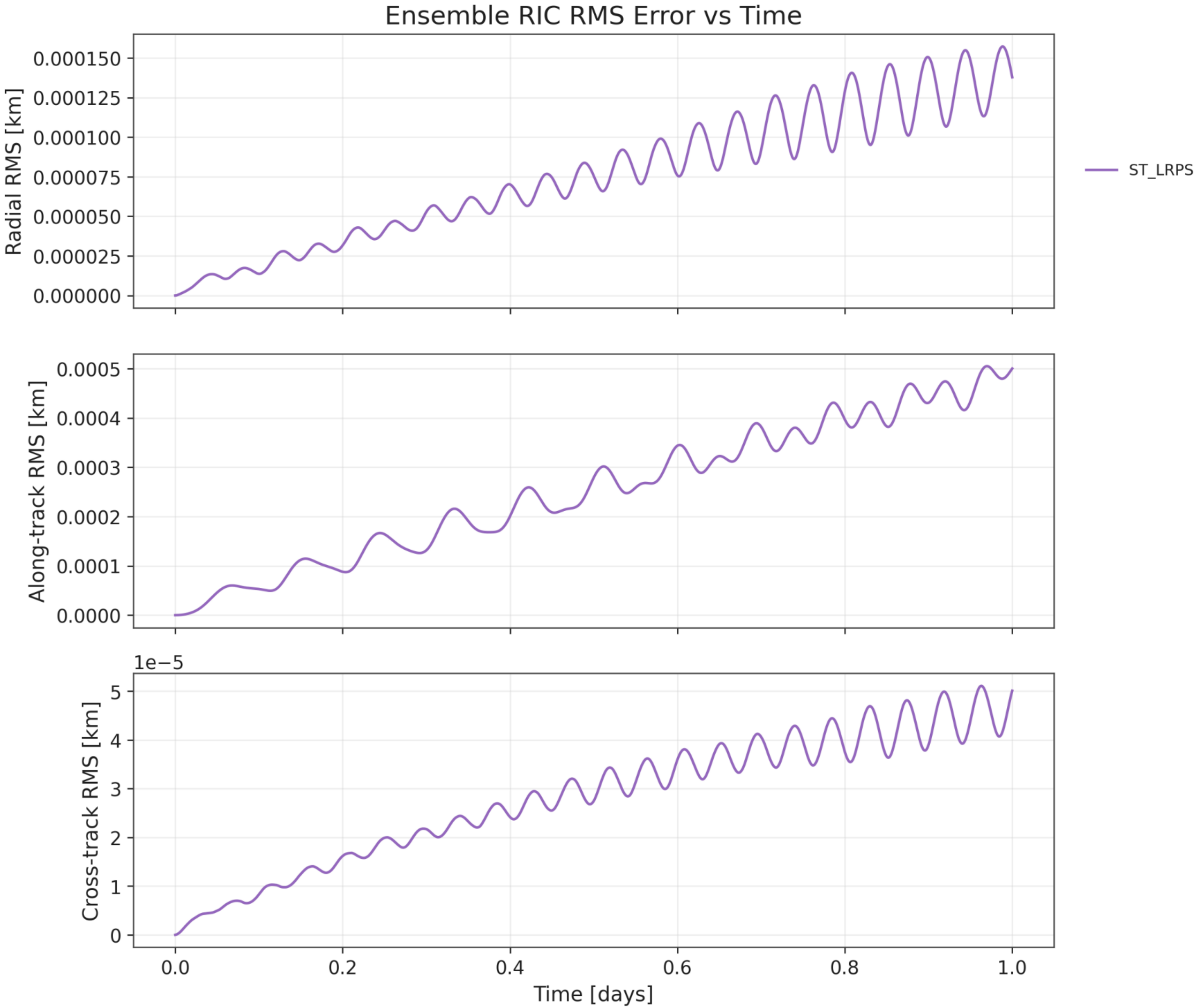


ST-LRPS Equivalent SH Degree
ST-LRPS median RMS
RMS Position Error [km]
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Spherical Harmonic Degree

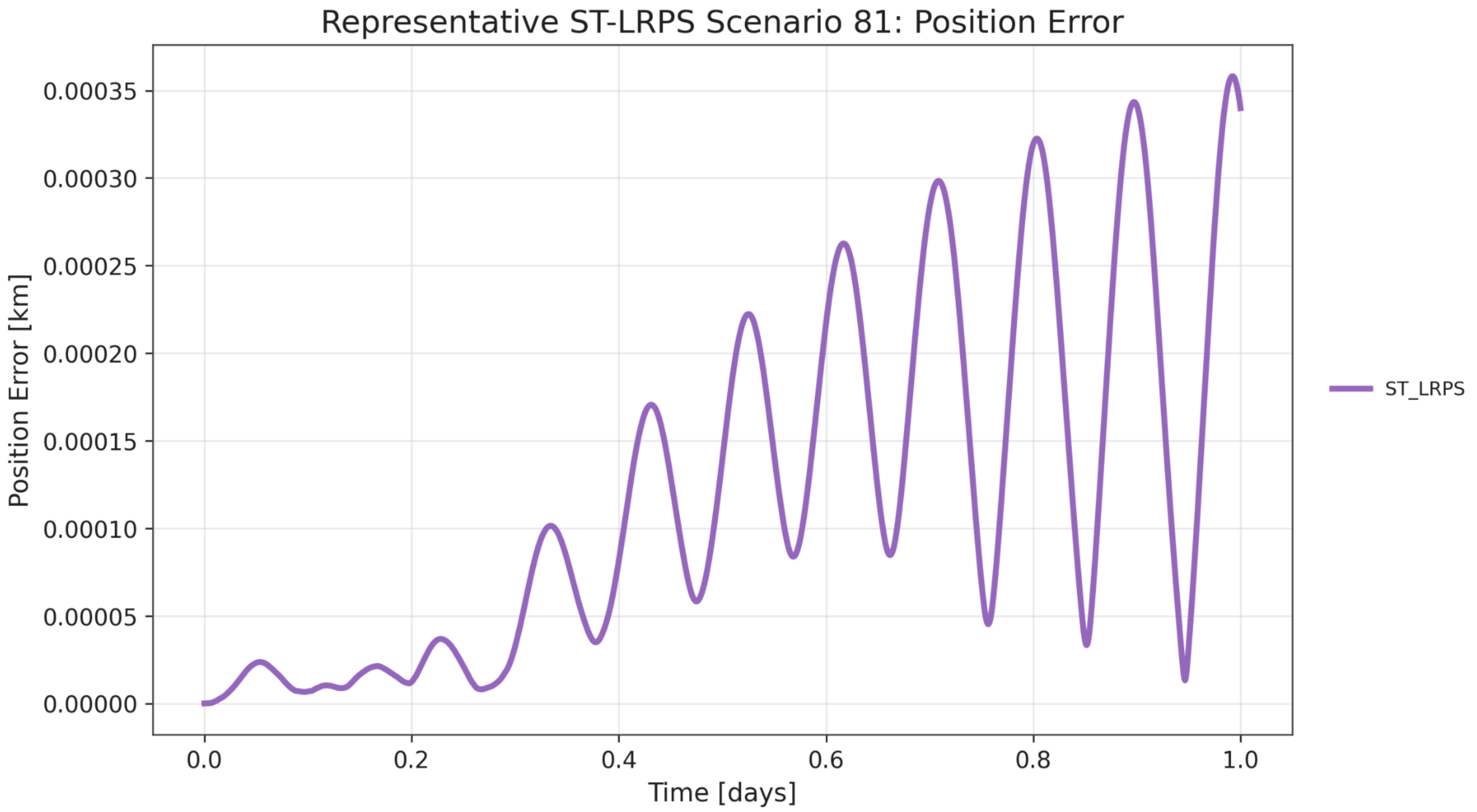

RMS Error Distribution Across Scenarios
RMS Position Error [km]
0.0005
0.0004
0.0003
0.0002
0.0001
GPU_ST_LRPS_RK4



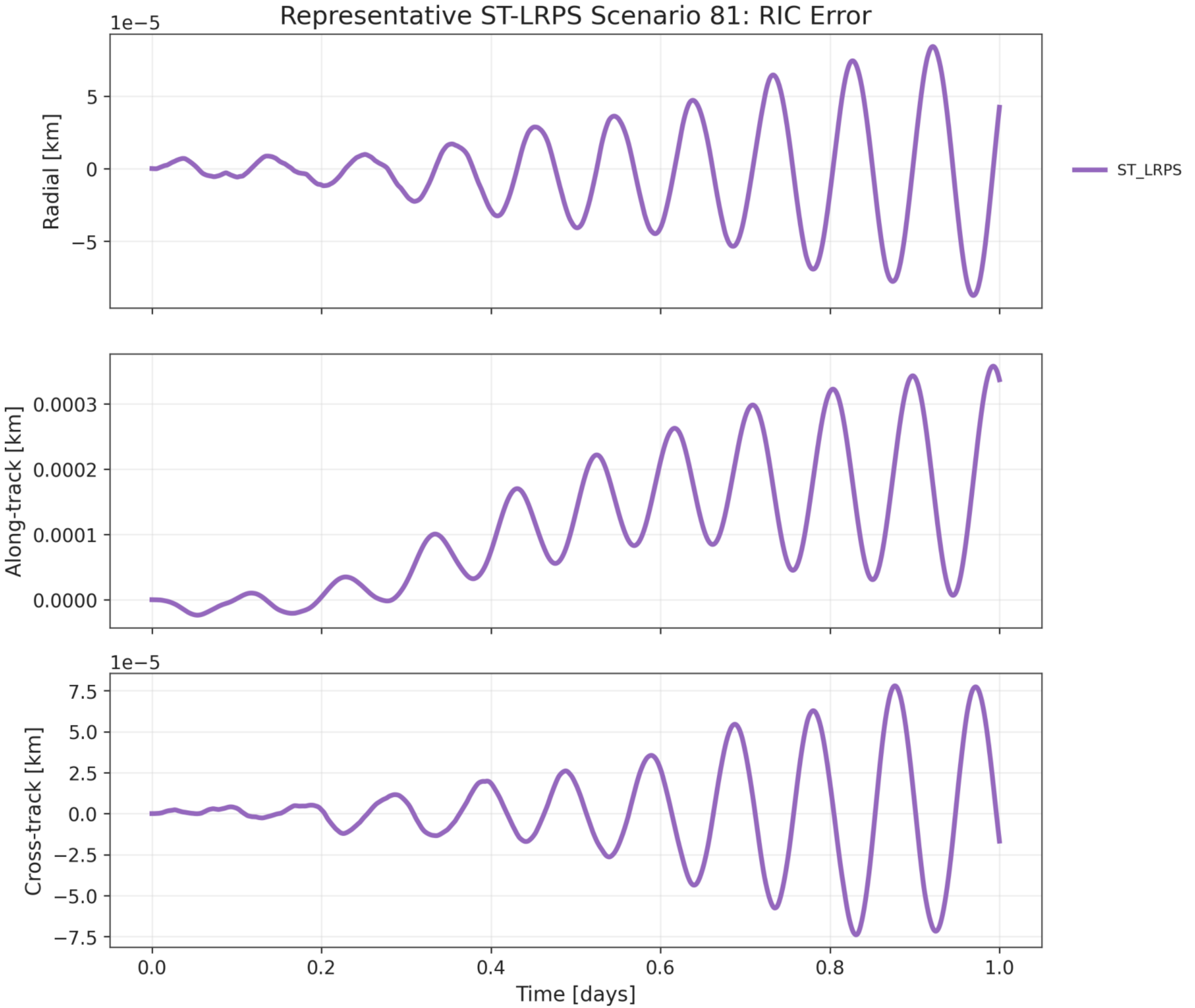
Ensemble Position Error vs Time
Median Position Error [km]
0.0000
0.0001
0.0002
0.0003
0.0004
0.0005
0.0
0.2
0.4
0.6
0.8
1.0
Time [days]
ST_LRPS



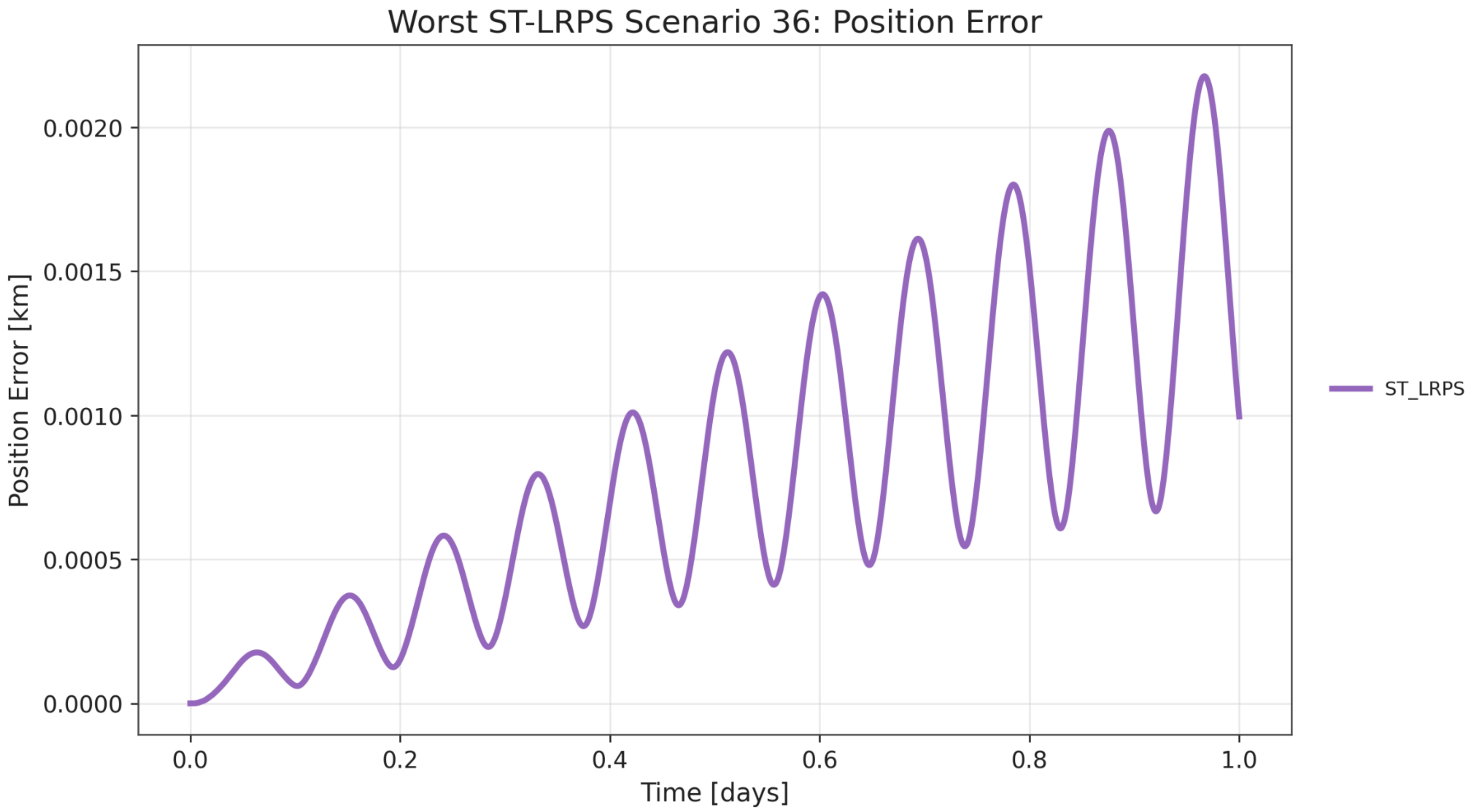
Ensemble RIC RMS Error vs Time
Radial RMS [km]
0.000000
0.000025
0.000050
0.000075
0.000100
0.000125
0.000150
ST_LRPS
Along-track RMS [km]
0.0000
0.0001
0.0002
0.0003
0.0004
0.0005
Cross-track RMS [km]
1e−5
0
1
2
3
4
5
0.0
0.2
0.4
0.6
0.8
1.0
Time [days]



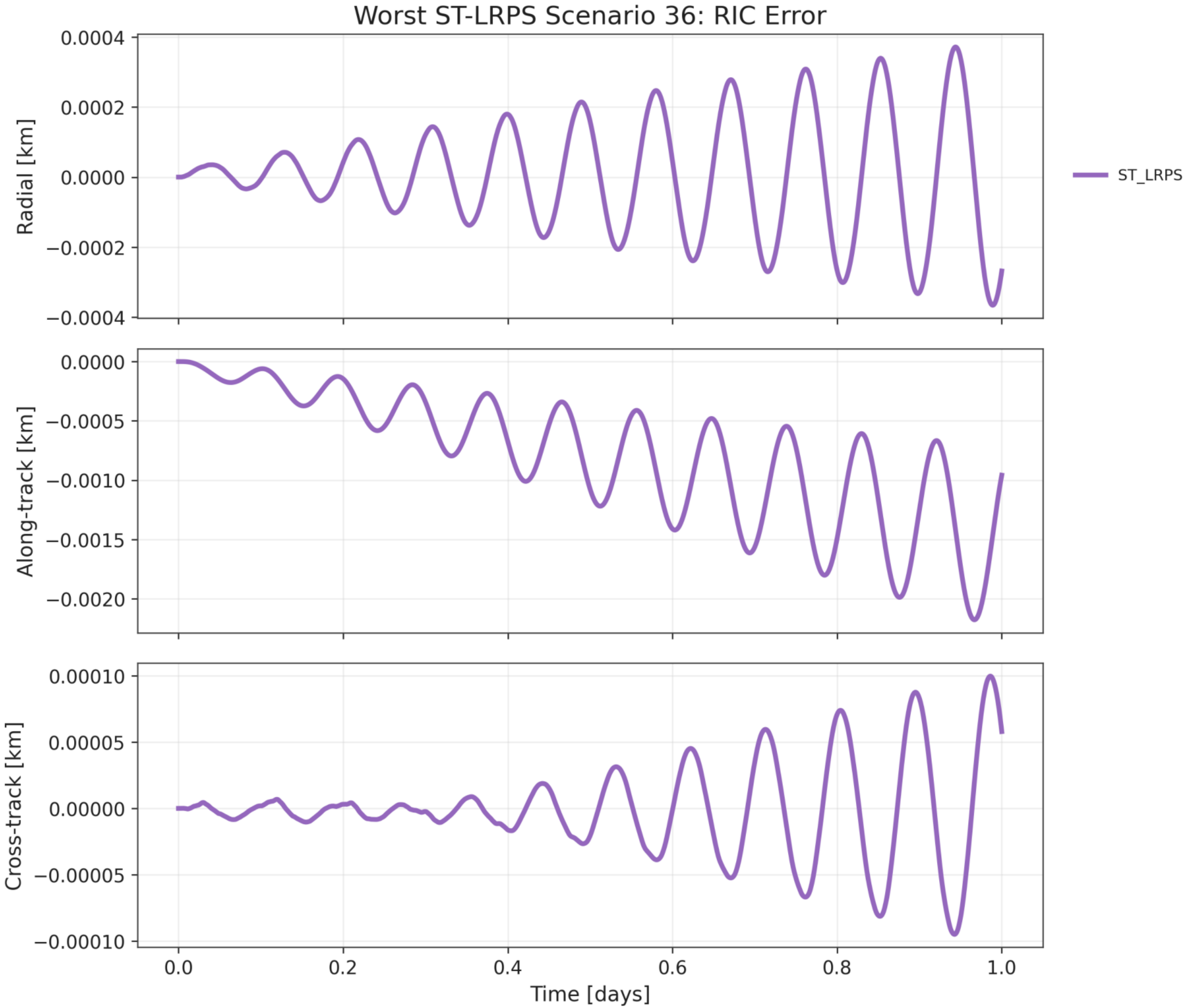
Representative ST-LRPS Scenario 81: Position Error
0.00035
0.00030
0.00025
0.00020
0.00015
0.00010
0.00005
0.00000
Position Error [km]
0.0
0.2
0.4
0.6
0.8
1.0
Time [days]
ST_LRPS



Representative ST-LRPS Scenario 81: RIC Error
1e−5
Radial [km]
5
0
−5
ST_LRPS
Along-track [km]
0.0003
0.0002
0.0001
0.0000
1e−5
Cross-track [km]
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
0.0
0.2
0.4
0.6
0.8
1.0
Time [days]



Worst ST-LRPS Scenario 36: Position Error
Position Error [km]
0.0000
0.0005
0.0010
0.0015
0.0020
Time [days]
0.0
0.2
0.4
0.6
0.8
1.0
ST_LRPS


Worst ST-LRPS Scenario 36: RIC Error
Radial [km]
0.0004
0.0002
0.0000
−0.0002
−0.0004
ST_LRPS
Along-track [km]
0.0000
−0.0005
−0.0010
−0.0015
−0.0020
Cross-track [km]
0.00010
0.00005
0.00000
−0.00005
−0.00010
0.0
0.2
0.4
0.6
0.8
1.0
Time [days]

Notes

```
Notes / warnings:
- SH200 DOP853 is a high-accuracy numerical reference, not physical ground truth.
- GPU SH200 RK4 vs SH200 DOP853 estimates fixed-step RK4 and framework error.
- Lower-degree GPU SH models include both truncation and RK4/framework error.
- ST-LRPS includes surrogate-model error plus RK4/framework error.
- Frame mode: match_dynamics_engine
```